

FIRETRACK

Incident Record Tracking and Management System for
BFP–Benguet

Bangonan, J.J.C. • Balubar, W.P. • Bala-oy, R.T. • Teofilo, L.L.B.
BS Information Technology | King's College of the Philippines | La Trinidad, Benguet

PROBLEM



Inefficient Record-Keeping

Paper logs, physical filing, and fragmented records across stations



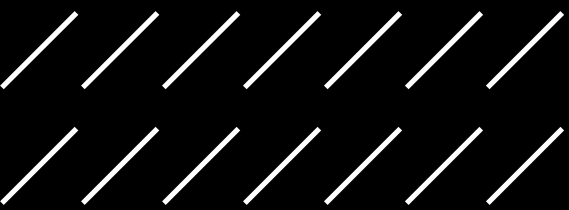
Fragmented Report Consolidation

Fire incident reports from the 13 municipalities are submitted separately via email



Inefficient Record-Keeping

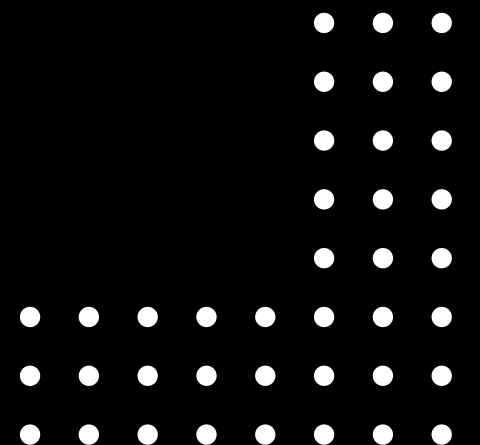
Absence of analytics, visualizations, or trend-monitoring tools

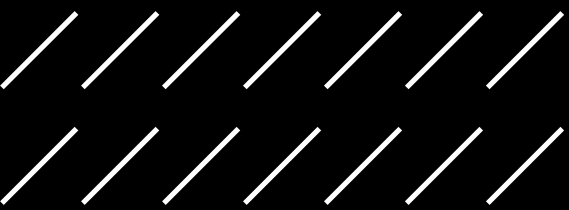


PROBLEM 1

MANUAL RECORD MANAGEMENT

According to Manun-og et al. (2022), implementing a records management system improves records organization, retrieval, and monitoring for government agencies. This supports the need to replace paper logs, physical filing, and fragmented records with a centralized digital records management system.

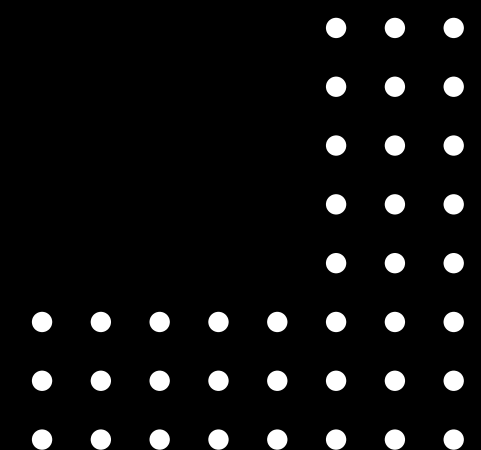


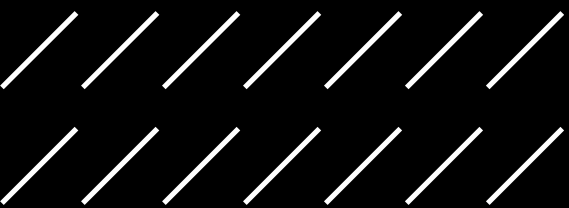


PROBLEM 2

FRAGMENTED REPORT CONSOLIDATION

Based on interviews and requirements-gathering activities conducted with personnel from the BFP–Benguet Provincial Office, fire incident reports from the 13 municipalities are submitted separately via email rather than through a centralized reporting system, requiring consolidation at the provincial office.

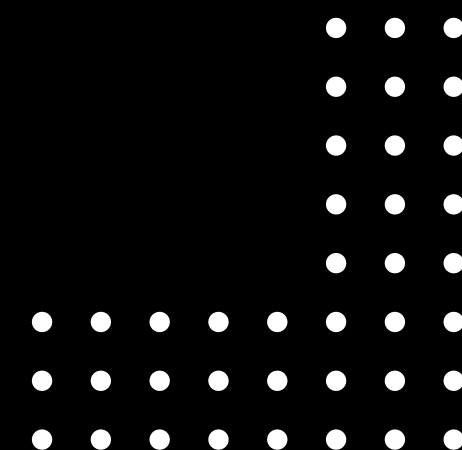




PROBLEM 3

LIMITED ANALYTICS AND VISUALIZATION

According to Asor et al. (2021), visualizing historical fire incident data and recognizing fire patterns can help identify temporal and geographic trends that support planning and decision-making. This highlights the need for dashboards and analytics in fire incident management systems.





SYSTEM FEATURES



6 Role-Based Accounts

- Fire Investigator
- Municipal Chief IIS
- Municipal Fire Marshal
- Special Operation
- Provincial Chief IIS
- Provincial Operations Section



Analytics Dashboard

- Monthly & yearly views
- Compare across years
- Municipality breakdown tables
- Charts & visualizations



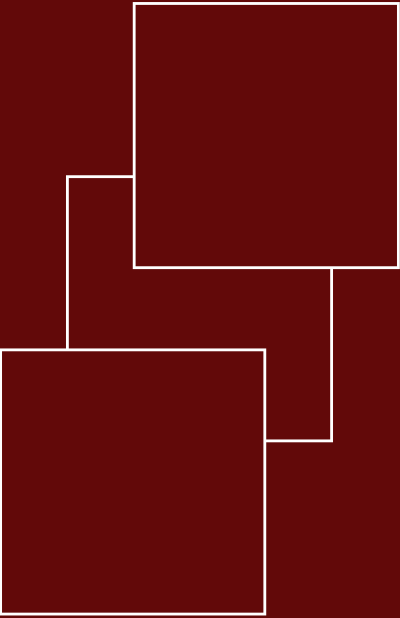
Structured Report Types

- Spot Report
- Progress Report
- Final Report
- MDFIR
- After Operation Report
- EMS Report



Workflow

- Approval / Return-for-correction trail
- BFP reference number formats
- PDF attachment uploads
- Role-based access control



DEVELOPMENT METHODOLOGY

Empathize

Semi-structured interviews & shadowing with BFP-Benguet personnel to map current workflows and pain points

Define

User personas, journey maps, and problem statement: "automated centralized data ecosystem for BFP-Benguet"

Ideate

Feature mapping, ERDs, system architecture, dynamic reporting interface, geospatial risk dashboard design

Prototype

Low-fidelity wireframes → High-fidelity full-stack web application using React, Node.js, PostgreSQL, Leaflet

Test

User acceptance testing with selected BFP personnel using ISO/IEC 25010 checklist and SUS questionnaire

THANK YOU

